

R.M.D. ENGINEERING COLLEGE

Smart India Hackathon 2026

PROJECT REPORT

AQUASHIELD

AI-Powered Sonar-Style Image Analysis Prototype

Problem Statement ID: SIH26057

AI-Powered Automated Underwater Marine Debris and Anomaly Detection Using Side-Scan Sonar Imagery

Category: Software

Team Name: Hackcult

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Live demo: <https://jesika-23-aquashield-app-bnkviv.streamlit.app/>
GitHub: https://github.com/Jesika-23/Hackcult_AquaShield

TABLE OF CONTENTS

Section.....	Page No.
1. Introduction.....	3
a) Project Overview.....	3
b) Problem Statement.....	3
c) Objectives.....	3
d) Target Users and Impact.....	3
e) Project Team.....	4
2. Problem Background and Existing Solutions.....	5
a) Current Practice and Challenges.....	5
b) Research Landscape.....	5
c) What Sets AquaShield Apart.....	5
3. Proposed Solution.....	6
a) Solution Overview.....	6
b) Detection and Analysis Workflow.....	6
c) System Architecture.....	6
d) Design Principle: Replaceable Model.....	7
4. System Modules and Technology Stack.....	8
a) Modules.....	8
b) Technology Stack.....	8
c) Features Implemented.....	8
d) Repository Structure.....	9
5. AI Model and Performance.....	10
a) Model Details.....	10
b) Training Results.....	10
c) Supporting Analysis Components.....	11
6. Development Journey: Beginning to Present.....	12
a) Phases of Work.....	12
b) Technical Challenge and Resolution.....	12
7. Compliance and Project Status.....	13
a) Mapping to Problem Statement SIH26057.....	13
b) Completed vs. Planned Checklist.....	13
c) Progress Statement.....	14
8. Deployment and Demonstration.....	15
a) Live Demo and Deployment.....	15
b) Local Setup.....	15
c) Demo Walkthrough.....	15
9. Limitations and Disclaimer.....	16
10. Future Work.....	16
11. Conclusion.....	17
12. References.....	17

1. Introduction

a) Project Overview

AquaShield is a research and educational prototype for AI-assisted analysis of underwater and sonar-style images. It uses a YOLO11n object-detection model to identify objects in uploaded images and presents the results through a Streamlit web application. Around the detector, it provides confidence filtering, image-quality assessment, prototype priority scoring, shadow-related heuristic cues, structured reporting, manual location support, detection history and human feedback logging.

AquaShield demonstrates how AI can support future marine-debris monitoring and seabed-survey workflows. It is designed as a decision-support prototype that assists human reviewers and does not replace expert judgment. It is not a production-ready or safety-certified navigation system.

Project status: Functional Streamlit research prototype, approximately 40% of the planned system. The core detection workflow is implemented; advanced deployment, marine-debris datasets, and GIS/field-validation modules are future work.

Live demo: <https://jesika-23-aquashield-app-bnkviv.streamlit.app/>

b) Problem Statement

Problem Statement ID: SIH26057

Title: AI-Powered Automated Underwater Marine Debris and Anomaly Detection Using Side-Scan Sonar Imagery

- Side-scan sonar and other underwater imaging produce pictures of the sea floor that can contain objects that are difficult and time-consuming to identify manually.
- Current review processes are slow when large numbers of images must be inspected, subjective across different analysts, and difficult to scale for large seabed survey areas.
- They are also limited in their ability to prioritise detections for further review.
- AI-based object detection with supporting analysis can help reviewers identify potential objects faster and organise their attention more effectively.

c) Objectives

- Use AI to review underwater and sonar-style images automatically instead of relying only on slow manual inspection.
- Detect objects and report a class label, bounding box and confidence score for each.
- Let the user control which detections are shown through an adjustable confidence threshold.
- Give reviewers supporting context through image-quality assessment, prototype priority scoring and shadow-related heuristic cues.
- Provide structured reports, manual location support, detection history and a feedback log.
- Build a foundation that can be extended with real marine-debris data and models in future.

d) Target Users and Impact

Marine debris, abandoned fishing gear, plastic waste, wreckage and other seabed hazards can affect marine ecosystems, fisheries and navigation safety. AquaShield is intended to support the following kinds of users in future marine-debris monitoring and survey workflows:

- Port authorities and harbour masters

- Hydrographic survey agencies and oceanographers
- Fisheries and coastal regulation departments
- Marine cleanup NGOs and research groups

The expected benefits are faster, AI-assisted screening of image collections, help with prioritising which detections to review first, and a documented record of analyses and reviewer feedback.

e) Project Team

Table 1.1: Team details

Item	Details
Team name	Hackcult
Institution	R.M.D. Engineering College
Problem Statement ID	SIH26057
Team lead	Haripriya S
Team members	Jesika M V, Kodavala Dhurgasree, Rupendra M, Pranav Mani S, Sanjay Raaj D
Department and year	Computer Science and Engineering, Year II
Mentor	Rajeshwari P

2. Problem Background and Existing Solutions

a) Current Practice and Challenges

Agencies may collect large volumes of side-scan sonar or underwater imagery. Reviewing these images manually is slow, subjective and hard to scale, and there is usually no automated help for deciding which detections deserve attention first. Three practical challenges shape the design of AquaShield:

- **Limited labelled debris data.** Underwater debris datasets with accurate labels are scarce, so the current model uses proxy classes.
- **Variable image quality.** Low contrast, noise, blur, poor visibility and uneven intensity affect detection reliability, so quality information is shown next to the detections.
- **Need for human judgment.** Predictions can contain false positives and false negatives, so results are framed as decision support for a human reviewer.

b) Research Landscape

Many research systems already do similar core work, and the team acknowledges this openly. The table below summarises the landscape and what it means for AquaShield.

Table 2.1: Existing work and its implication for this project

Area	What already exists	Implication for AquaShield
YOLO-based side-scan detection	Many papers train YOLOv5, YOLOv7 or YOLOv8 on side-scan sonar images to find shipwrecks, small targets and objects in complex environments.	Training YOLO on sonar-style images is not novel by itself, and AquaShield does not claim it is.
Acoustic-shadow analysis	An established technique in the research literature.	AquaShield uses a heuristic computer-vision cue and does not claim a validated acoustic-shadow model.
Deep-learning debris detection	Marine debris detection with deep learning exists, for example work on forward-looking sonar debris datasets.	AI debris detection is an established idea. Labelled side-scan debris data is the missing piece.
Typical published systems	A model and its metrics reported in a paper.	AquaShield packages detection with supporting analysis in a working application.

c) What Sets AquaShield Apart

The core YOLO detection is not the novelty. The difference lies in how the solution is packaged and used:

- **Detection plus supporting analysis.** Results come with image-quality context, a prototype priority level and shadow-related cues, not only bounding boxes.
- **Quality awareness.** Image-quality information is shown alongside detections so reviewers can judge how far to trust them.
- **Reporting and traceability.** Structured reports, detection history and manual location support document each analysis.
- **Feedback logging.** Reviewer feedback is stored in a CSV file as a foundation for labelling, error analysis and future model improvement.
- **Honest scope.** The proxy classes, heuristics and prototype status are stated clearly throughout.

3. Proposed Solution

a) Solution Overview

The user uploads an underwater or sonar-style image in the Streamlit application. The image is preprocessed and assessed for quality, and the YOLO11n model then detects objects and draws a bounding box with a class label and confidence score around each one. Only detections that meet the selected confidence threshold (default 0.50) are shown. The application then adds a prototype priority level and shadow-related cues, lets the user generate a structured report, add a manual location and give feedback, and records the analysis in the detection history.

Important honest point: the model currently detects four proxy classes (Aircraft, Fish, Reef and Ship), not real marine debris. The classes are used to demonstrate the object-detection pipeline, and future versions will need training on labelled marine-debris and real sonar datasets.

b) Detection and Analysis Workflow

Table 3.1: Processing steps

Step	Action
1	Upload an underwater or sonar-style image.
2	Prepare the image using OpenCV preprocessing.
3	Assess image quality, which can affect detection reliability.
4	Run the YOLO11n model to identify potential objects.
5	Filter detections using the confidence threshold (default 0.50).
6	Display bounding boxes, classes and confidence scores on the processed image.
7	Show supporting analysis: image quality, prototype priority level and shadow-related cues.
8	Generate a structured report summarising the analysis.
9	Optionally enter coordinates manually and view the location on a map.
10	Review detection history and record feedback on detections.

c) System Architecture

The current implementation is an integrated Streamlit application. The interface, preprocessing, model inference, supporting analysis, reporting, history, feedback and map components all run within one application.

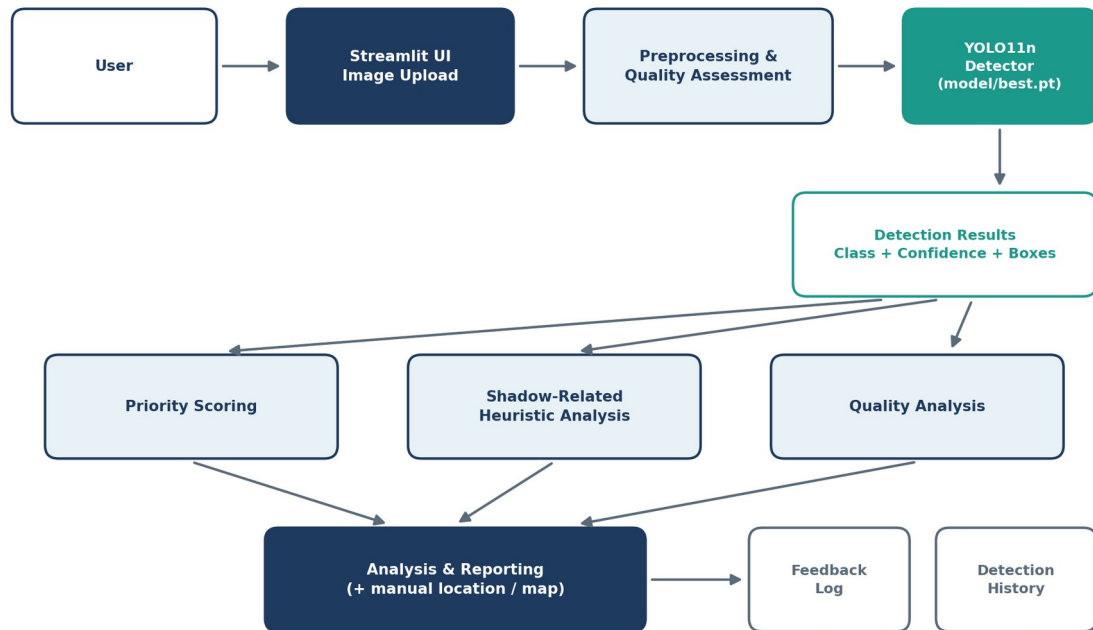


Figure 3.1: Current AquaShield system architecture (integrated Streamlit prototype)

d) Design Principle: Replaceable Model

The detector is loaded from a single weights file (model/best.pt) through a separate inference module, and the supporting analysis components work on generic detection results (class, confidence and bounding box). This modular structure is intended to make it easier to integrate a model trained on real marine-debris data in future.

4. System Modules and Technology Stack

a) Modules

Table 4.1: Modules and their locations

Module	What it does	Where it lives
Web application (Streamlit)	Upload, controls, results display and integration of all components	app.py
Object detection (YOLO11n)	Runs the trained model and returns class, bounding box and confidence	utils/inference.py, model/best.pt
Preprocessing	Prepares the uploaded image for analysis using OpenCV	utils/preprocess.py
Image quality assessment	Evaluates image characteristics that may affect detection reliability	utils/quality.py
Priority scoring	Assigns a prototype priority level to detections	utils/priority.py
Shadow-related analysis	Estimates visual characteristics around detected objects (heuristic)	utils/shadow.py
Report generation	Builds a structured analysis summary	utils/report.py
Detection history	Tracks previous analysis activity	utils/history.py
Feedback logging	Stores human feedback on detections	utils/feedback.py, feedback_log.csv
Map and location support	Shows manually entered coordinates on a map	utils/map_view.py
Styling	Look and feel of the application	assets/style.css

b) Technology Stack

Table 4.2: Technologies used

Technology	Purpose
Python 3.12 (recommended)	Core application and AI processing. Python 3.13 caused errors with PyTorch, so 3.12 is recommended.
Streamlit	Web application interface
Ultralytics YOLO11n	Object detection
OpenCV	Image preprocessing and computer vision
Pandas	Data handling and feedback logging
Pillow	Image loading and processing
Plotly	Data visualisation
Streamlit Community Cloud	Deployment of the application, connected to the GitHub repository

c) Features Implemented

- Upload of underwater or sonar-style images
- Image preprocessing using OpenCV

- YOLO11n object detection with bounding-box visualisation, class and confidence
- Adjustable confidence threshold (default 0.50)
- Image-quality assessment
- Prototype priority scoring with levels HIGH, REQUIRES HUMAN VERIFICATION, MEDIUM and LOW
- Shadow-related heuristic analysis
- Structured report generation
- Manual map and location support
- Detection history
- Human feedback logging to feedback_log.csv

d) Repository Structure

```
Hackcult_AquaShield/
├── app.py
├── requirements.txt
├── feedback_log.csv
├── .gitignore
├── README.md
├── model/
│   └── best.pt
├── assets/
│   └── style.css
├── utils/
│   ├── __init__.py
│   ├── inference.py
│   ├── preprocess.py
│   ├── quality.py
│   ├── priority.py
│   ├── shadow.py
│   ├── report.py
│   ├── history.py
│   ├── feedback.py
│   └── map_view.py
└── text/
    └── images/
```

5. AI Model and Performance

a) Model Details

Table 5.1: Model summary

Item	Details
Model type	YOLO11n (Ultralytics), a small and fast object detector
Weights file	model/best.pt
Training images	402
Validation images	110
Default confidence threshold	0.50 (adjustable by the user)
Nature of the classes	Prototype (proxy) classes used to demonstrate the detection pipeline. They are not final marine-debris categories.

Table 5.2: Prototype detection classes

Class ID	Prototype class
0	Aircraft
1	Fish
2	Reef
3	Ship

These classes are used only to demonstrate the object-detection pipeline. Future versions will require training on labelled marine-debris and real sonar datasets for classes such as ghost nets, plastic debris, wreckage and other seabed hazards.

b) Training Results

The metrics below were observed during model development on the current prototype dataset and are reported as obtained.

Table 5.3: Model performance

Metric	Value	Meaning
Precision	47.3%	Share of the detections made that are correct
Recall	55.3%	Share of the real objects that the model finds
mAP@0.50	51.5%	Mean average precision at an IoU threshold of 0.5
mAP@0.50-0.95	31.4%	Mean average precision averaged over IoU thresholds from 0.5 to 0.95

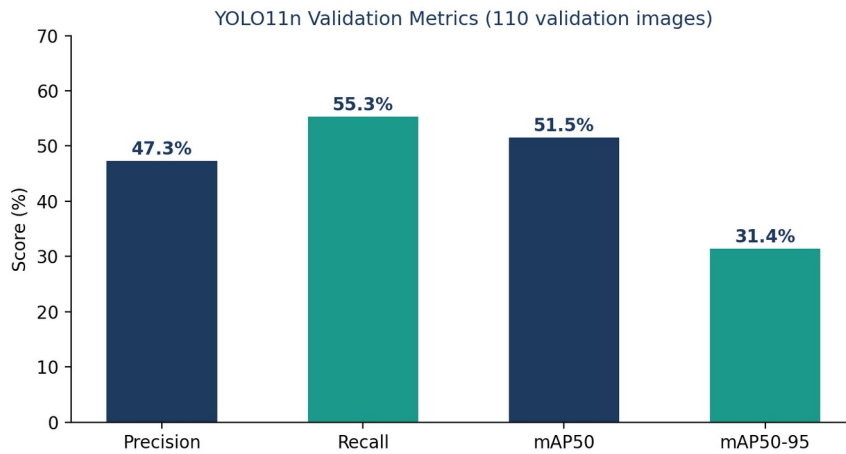


Figure 5.1: YOLO11n performance metrics

These figures should not be interpreted as production-level performance or as final marine-debris detection accuracy. The model was trained on a relatively small dataset with proxy classes, so predictions may contain false positives and false negatives.

c) Supporting Analysis Components

Around the detector, AquaShield provides supporting components that give a human reviewer additional context. These components are classical computer-vision or rule-based prototype heuristics and not scientifically validated measurements.

Table 5.4: Supporting analysis components

Component	How it works	Status
Image quality assessment	Evaluates characteristics such as low contrast, noise, blur, poor visibility and uneven intensity that can affect detection reliability, and shows quality information next to the detections.	Prototype heuristic
Prototype priority scoring	Helps reviewers identify detections that may need more attention, using signals such as detection confidence, object size, image quality and shadow-related cues. Levels: HIGH, REQUIRES HUMAN VERIFICATION, MEDIUM, LOW.	Prototype heuristic, for decision support only
Shadow-related analysis	Estimates visual characteristics around detected objects and uses them as an additional supporting signal.	Heuristic; not a validated sonar-physics or acoustic-shadow model
Structured report	Summarises the image analysed, detected objects, confidence values, priority, image-quality information and supporting results.	Implemented
Feedback logging	Users review detections and give feedback, which is stored in feedback_log.csv.	Stored for future data labelling and model improvement; the model does not learn from it yet

6. Development Journey: Beginning to Present

a) Phases of Work

The phases below summarise the work recorded in the project documentation.

Table 6.1: Development phases

Phase	Work carried out	Outcome
1. Problem study and ideation	Studied problem statement SIH26057, reviewed related research and defined the solution idea. Prepared the Smart India Hackathon idea presentation.	Solution concept and workflow design
2. Model training	Trained YOLO11n on 402 images and validated it on 110 images, using proxy classes (Aircraft, Fish, Reef, Ship) because labelled marine-debris sonar data was not available.	model/best.pt with 47.3% precision, 55.3% recall and 51.5% mAP@0.50
3. Streamlit prototype	Built the Streamlit application with image upload, preprocessing, YOLO detection, bounding-box visualisation and the confidence threshold.	Working core detection workflow
4. Supporting analysis	Added image-quality assessment, prototype priority scoring, shadow-related heuristic analysis, structured reporting, manual map and location support, detection history and feedback logging.	Detector extended into a decision-support prototype
5. Environment stabilisation	Fixed a start-up failure caused by Python 3.13 and PyTorch (see below).	Stable runtime on Python 3.12
6. Cloud deployment of the prototype	Deployed the integrated Streamlit application on Streamlit Community Cloud, connected to the GitHub repository.	Public live demo
7. Documentation and scope review	Documented features, limitations and disclaimers, and separated completed work from planned future extensions.	Clear completed-versus-planned status

b) Technical Challenge and Resolution

The application failed to start because Python 3.13 produced errors with PyTorch. The team resolved this by:

- Deleting the old, broken virtual environment
- Creating a fresh virtual environment with Python 3.12
- Installing the required packages

The application then ran successfully, and Python 3.12 is now the recommended environment.

7. Compliance and Project Status

a) Mapping to Problem Statement SIH26057

Table 7.1: Problem statement requirements and current status

Requirement	Current implementation	Status
Automated AI-based review of images	YOLO11n detects objects automatically in uploaded underwater or sonar-style images	Implemented
Confidence scoring and noise filtering	Confidence score per detection, adjustable confidence threshold and image-quality assessment	Implemented
Acoustic-shadow and texture analysis	Shadow-related heuristic cues (not a validated acoustic-shadow model)	Partial
Geo-tagging and localisation	Manual coordinate entry with map display; no automatic geo-referencing	Partial
Visualisation, dashboard and reporting	Streamlit dashboard with bounding boxes, structured report, history and feedback log	Implemented
Anomaly detection	No dedicated anomaly-detection module in the current version; advanced anomaly detection is planned	Planned
Marine-debris classification	Model detects proxy classes only; needs labelled marine-debris sonar data	Planned

b) Completed vs. Planned Checklist

The table below separates what is working in the current prototype from what is planned as future work.

Table 7.2: Completed and planned items

Item	Status
Upload of underwater or sonar-style images	Completed
Image preprocessing using OpenCV	Completed
YOLO11n object detection	Completed
Bounding-box visualisation	Completed
Confidence-threshold filtering	Completed
Image-quality assessment	Completed
Prototype priority scoring	Completed
Shadow-related heuristic analysis	Completed
Structured report generation	Completed
Manual map and location support	Completed
Detection history	Completed
Human feedback logging	Completed
Deployment as an integrated Streamlit application on Streamlit Community Cloud	Completed
Training on real, labelled marine-debris sonar datasets	Planned

Item	Status
Marine-debris-specific classes (ghost nets, plastic debris, wreckage, seabed hazards)	Planned
Advanced anomaly detection	Planned
Separation into a FastAPI backend and Streamlit frontend	Planned
Dockerised deployment on cloud (for example AWS) and edge infrastructure	Planned
Real geo-referencing using survey navigation data	Planned
Debris clustering and prioritised debris maps	Planned
Navigation-risk and ecological-sensitivity GIS layers	Planned
ONNX Runtime optimisation for lightweight CPU inference	Planned
Field validation with hydrographic survey agencies, port authorities and marine-cleanup organisations	Planned

c) Progress Statement

Status: Functional Streamlit prototype, approximately 40% of the planned system.

Functional Streamlit research prototype with core detection workflow implemented; advanced deployment, marine-debris datasets, and GIS/field-validation modules are future work. The completed items above make up the working prototype, and the planned items make up the remaining roadmap described in Section 10.

8. Deployment and Demonstration

a) Live Demo and Deployment

- The application is deployed on Streamlit Community Cloud and connected to the GitHub repository.
- The deployed application installs its dependencies from the project's requirements.txt file.
- Live demo: <https://jesika-23-aquashield-app-bnkviv.streamlit.app/>
- GitHub repository: https://github.com/Jesika-23/Hackcult_AquaShield
- The public deployment is the integrated Streamlit prototype. Separating the system into a FastAPI backend and Streamlit frontend, with Docker-based deployment on cloud platforms such as AWS or on edge infrastructure, is planned future work.

b) Local Setup

Prerequisites: Python 3.12 (recommended) and a virtual environment.

Steps:

1. Clone the repository:

```
git clone https://github.com/Jesika-23/Hackcult_AquaShield.git
```

```
cd Hackcult_AquaShield
```
2. Create a virtual environment:

```
python -m venv venv
```
3. Activate it:

```
venv\Scripts\activate
```

 (Windows)

```
source venv/bin/activate
```

 (Linux / macOS)
4. Install dependencies:

```
pip install -r requirements.txt
```
5. Run the application:

```
streamlit run app.py
```

The application opens in your browser.

c) Demo Walkthrough

- Open the live demo link and upload an underwater or sonar-style image.
- Adjust the confidence threshold if needed (default 0.50) and analyse the image.
- Show the bounding boxes with class names and confidence scores on the processed image.
- Point out the image-quality information, the prototype priority level and the shadow-related cues.
- Generate the structured report.
- Enter coordinates manually to show the map and location support.
- Show the detection history and record feedback on a detection.
- Backup plan: if the live demo is unavailable, use a recorded video or screenshots.

9. Limitations and Disclaimer

AquaShield is a **research and educational prototype**. The current detection model was trained on a relatively small dataset and uses prototype or proxy object classes. Predictions may contain false positives or false negatives.

The image-quality assessment, priority scoring, map and location support, and shadow-related analysis are prototype heuristic components. They should be treated as supporting decision-making features and not as scientifically validated sonar measurements. In particular:

- The shadow-related component is **not** a validated acoustic-sonar physics model.
- The current object classes (Aircraft, Fish, Reef, Ship) are **not** final marine-debris categories.
- The model is **not** validated on operational marine-debris survey data.
- Location support is manual: users type coordinates and there is no automatic geo-referencing.
- Feedback is logged for future use, but the model does not learn from it yet.
- The system is **not** certified for navigation or safety-critical operations.
- All detections must be reviewed and verified by a qualified human operator before any real-world action is taken.

10. Future Work

The current prototype represents approximately 40% of the planned system. The extensions below make up the remaining roadmap, and none of them is part of the current repository version.

Table 10.1: Planned enhancements

Enhancement	What will be done	Limitation addressed
Real marine-debris datasets and classes	Train on larger, labelled marine-debris and real side-scan sonar datasets with classes such as ghost nets, plastic debris, wreckage and seabed hazards.	Proxy classes and small dataset
Advanced anomaly detection	Add dedicated anomaly detection for unusual objects that do not fit normal patterns.	No dedicated anomaly module
Validated acoustic-shadow modelling	Improve the shadow module using real side-scan sonar data and validated acoustic-shadow models.	Heuristic shadow cues
Real geo-referencing	Integrate GPS, survey-mission, vessel and navigation data instead of manual coordinates.	Manual location entry
Debris clustering and prioritised maps	Cluster detections and produce prioritised debris maps.	No spatial prioritisation
GIS risk layers	Add navigation-risk and ecological-sensitivity layers to the priority score.	Priority based only on image-level signals
FastAPI backend and Streamlit frontend	Separate the system into a backend service and a frontend.	Single integrated application
Dockerised cloud and edge deployment	Package the system in Docker and deploy it on cloud platforms such as AWS, and on edge infrastructure.	Streamlit-only deployment

Enhancement	What will be done	Limitation addressed
ONNX Runtime optimisation	Optimise for lightweight CPU inference.	Limited compute on vessels
Improved detection accuracy	Use larger and more diverse real sonar datasets and further training to improve object-detection accuracy.	Moderate accuracy
Real side-scan sonar integration	Work directly with real side-scan sonar data from surveys.	Sonar-style images and proxy data
Multi-frame object tracking	Track objects across consecutive frames or survey lines.	Single-image analysis only
Cloud-based detection history	Store detection history persistently in the cloud.	History limited to the current session
Advanced mapping and monitoring dashboards	Extend the map view into richer mapping and monitoring dashboards.	Basic manual map view
Feedback-based model improvement	Use logged reviewer feedback for data labelling, error analysis and retraining.	Feedback not used for learning yet
Field validation	Validate with hydrographic survey agencies, port authorities and marine-cleanup organisations.	No operational validation

11. Conclusion

AquaShield shows how AI-assisted analysis of underwater and sonar-style images can be delivered as a working decision-support prototype and not only as a trained model. It combines YOLO11n object detection with image-quality assessment, prototype priority scoring, shadow-related heuristic cues, structured reporting, manual location support, detection history and feedback logging, and it is available as a live Streamlit application. It represents approximately 40% of the planned system.

The project is honest about its scope. The model was trained on a small dataset with proxy classes, its accuracy is moderate, and several supporting components are heuristics. The main next steps are training on labelled marine-debris and real sonar data, real geo-referencing, and the deployment, GIS and field-validation extensions listed in the completed-versus-planned checklist.

In one sentence: AquaShield is a functional Streamlit research prototype (approximately 40% of the planned system) with the core detection workflow implemented; advanced deployment, marine-debris datasets, and GIS/field-validation modules are future work.

12. References

Research papers

- YOLO-Based Sonar Object Detection, MDPI Sensors, 2022.
- Deep Learning for Underwater Target Recognition in Side-Scan Sonar, IEEE Access, 2023.
- Marine Debris Risk Assessment Frameworks, Elsevier, 2023.

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Articles

- Ocean Conservancy, Ghost Gear and Seabed Debris, 2023 to 2024.
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Tools and software

- Ultralytics YOLO11n, OpenCV, Streamlit, Pandas, Pillow, Plotly, PyTorch and Streamlit Community Cloud.